Supakorn Surarat

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Professional Summary

Computer Engineering undergraduate student at HKU pursuing an integrated BEng in Computer Engineering and MSc in AI in Engineering on a full Belt and Road Scholarship. Skilled in Python, C/C++, SQL, Linux, computer vision, and machine learning. Seeking a summer internship across Software Engineering, IT, AI/Data, or Computer Engineering.

Education

The University of Hong Kong (HKU) — Hong Kong SAR
Integrated BEng in Computer Engineering + MSc in AI in Engineering (Minor in Finance) — *Sept 2025 – Present*

- **Academic Standing**: CGPA 3.35 (Semester 2 GPA: 3.53) | Distinction in AI Literacy I (AILT1001)
- **Coursework**: Data Structures & Algorithms, C/C++ Programming, Python Programming, Microprocessors, Electricity & Electronics (B+), Linear Algebra & Probability

Princess Chulabhorn Science High School Pathum Thani — Pathum Thani, Thailand
High School Diploma (Science, Technology & Applied Research Curriculum) — *2022 – 2025*

- **Highlights**: Cumulative GPAX 3.96 | Exchange Program to USA (Marcos de Niza High School, Arizona)

Technical Skills

Programming & Scripting: Python, C/C++, SQL, Bash/Shell
AI, Computer Vision & Data: OpenCV, YOLO, Supervised Machine Learning, Google Earth Engine, Jupyter, Excel
Systems, Hardware & Tools: Linux, Raspberry Pi, Microprocessors, Git, GitHub, VS Code, CAD, 3D Printing
Languages: Thai (Native), English (Fluent — IELTS 7.5), Cantonese (Beginner)

Experience — Software, AI & Computer Engineering Projects

Autonomous Closed-Loop Target Tracking System — 2025 – 2026
Core Developer | Python, OpenCV, YOLO, Embedded Linux, Raspberry Pi — *HKU ENGG1101 Capstone (Grade A+)*

- Built a real-time computer vision system on a Linux-based Raspberry Pi using OpenCV and YOLO to detect human targets and extract bounding-box coordinates from live video streams.
- Programmed closed-loop feedback control in Python to calculate spatial offset errors and drive dual-axis servos, automatically locking onto moving targets in real time.

Satellite Image Processing & Machine Learning Classifier — 2023 – 2024
Co-Developer | Google Earth Engine, Landsat 8 Imagery, Supervised ML, NDVI — *National Showcase*

- Developed a supervised ML classifier in Google Earth Engine using Landsat 8 satellite imagery across six Thai coastal provinces.
- Engineered NDVI vegetation reflectance features to separate mangrove forests from inland forests, achieving **96.38%–99.70% accuracy** in Samut Songkhram (2014–2020).
- Selected by the Ministry of Education to present at the **Thailand–Japan Student ICT Fair 2024**.

Scholarships & Awards

- **HKSAR Government Belt and Road Scholarship** (2025): Full-tuition merit award for top international students.
- **HKSAR Government Talent Development Scholarship** (2025): Awarded for Innovation, Science & Technology.
- **HKU Scholarship for Future Leaders & CAES1001 Certificate of Excellence** (2025): Academic & leadership honors.
- **Chevron REACH Scholarship** (2025, 2026): Academic merit scholarship (*Recognizing Excellence and Achievement*).
- **International Invention Gold Medals** (2024): Best Innovation Award & Gold Medal at Indonesia Inventors Day (KREAD-TISOL Diagnostic Kit); Gold Medal & INNOPA Special Award at JDIE Tokyo, Japan (Aphid 3D-Printed Trap).
- **National Science & Olympiad Honors**: Gold Medal at 4th PCSHS Science Symposium 2024; Best Oral Presentation at 2nd Silapakorn Young Scientist Forum; POSN Astronomy Olympiad Camp 1.

Leadership & Extracurricular Activities

HKU Thai Student Association (ThaiSA) — *Event Director* — Hong Kong SAR | *April 2026 – Present*

- Direct cultural and orientation events for 50+ Thai students at HKU, coordinating venue logistics and 15+ volunteers.

Chi Sun College Badminton Club, HKU — *Team Captain* — Hong Kong SAR | *2026 – Present*

- Lead weekly college training sessions and co-organized the JCSVIII Inter-Badminton Tournament (brackets & court schedules).

HKU Faculty of Engineering — *Faculty Student Ambassador* — Hong Kong SAR | *Sept 2025 – Present*

- Represented the BEng + MSc AIE program at the HKU Information Day main booth for prospective students and parents.